

Create_VM_Scale_sets

You can implement Azure Virtual Machine Scale Sets in the Azure portal. You specify the number of virtual machines and their sizes, and indicate preferences for using Azure Spot instances, Azure managed disks, and allocation policies.

In the Azure portal, there are several settings to configure to create an Azure Virtual Machine Scale Sets implementation.

Create a virtual machine scale set

BasicsDisksNetworkingScalingManagementHealthAdvancedTagsReview + create

Azure virtual machine scale sets let you create and manage a group of load balanced VMs. The number of VM instances can automatically increase or decrease in response to demand or a defined schedule. Scale sets provide high availability to your applications, and allow you to centrally manage, configure, and update a large number of VMs.

Orchestration

A scale set has a "scale set model" that defines the attributes of virtual machine instances (size, number of data disks, etc). As the number of instances in the scale set changes, new instances are added based on the scale set model.

Orchestration mode * ⓘ

☒ **Uniform:** optimized for large scale stateless workloads with identical instances

☐ **Flexible:** achieve high availability at scale with identical or multiple virtual

Instance details

Image * ⓘ

Ubuntu Server 20.04 LTS - x64 Gen2

[See all images](#) | [Configure VM generation](#)

VM architecture ⓘ

☐ Arm64

☒ x64

Run with Azure Spot discount ⓘ

☐

Size * ⓘ

Standard_D2s_v3 - 2 vcpus, 8 GiB memory

[See all sizes](#)

- **Orchestration mode:** Choose how the scale set manages the virtual machines. In flexible orchestration mode, you manually create and add a virtual machine of any configuration to the scale set. In uniform orchestration mode, you define a virtual machine model and Azure generates identical instances based on that model.
- **Image:** Choose the base operating system or application for the VM.
- **VM Architecture:** Azure provides a choice of x64 or Arm64-based virtual machines to run your applications. x64-based VMs provide the most software compatibility. Arm64-based VMs provide up to 50% better price-performance than comparable x64 VMs.
- **Size:** Select a VM size to support the workload that you want to run. The size that you choose then determines factors such as processing power,

memory, and storage capacity. Azure offers a wide variety of sizes to support many types of uses. Azure charges an hourly price based on the VM's size and operating system.

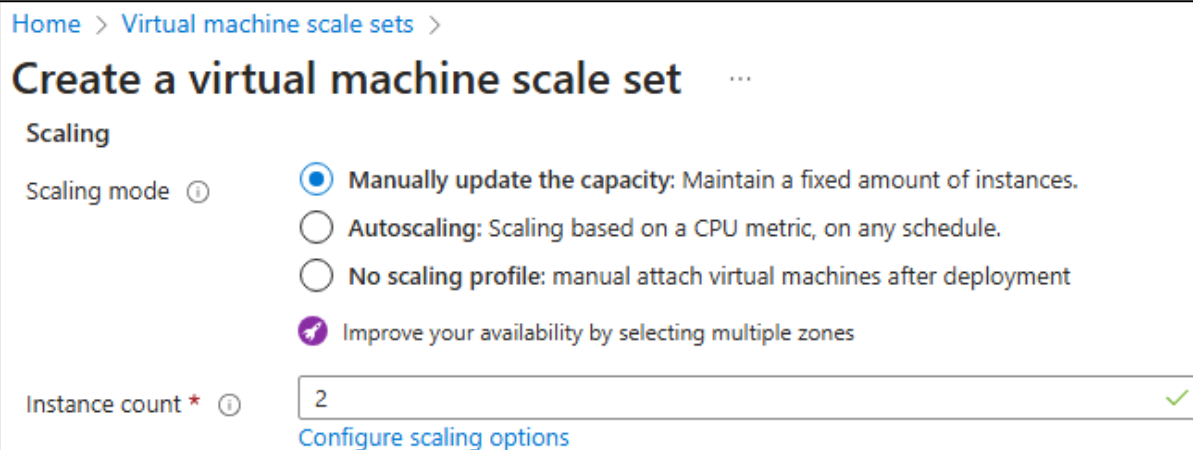
Under the Advanced tab, you can also select the following:

- Spreading algorithm: The spreading algorithm determines how VMs in the scale set are balanced across fault domains. With max spreading, VMs are spread across as many fault domains as possible in each zone. With fixed spreading, VMs are always spread across exactly five fault domains. In the case where fewer than five fault domains are available, a scale set using "Max spreading" completes, while a scale set using "Fixed spreading" fails. For this reason, Microsoft recommends using Max spreading for your implementation.

Implement auto scale

When you create an Azure Virtual Machine Scale Sets implementation in the Azure portal, you can enable manual or autoscaling. For optimal performance, you should define a minimum, maximum, and default number of virtual machine instances to use.

In the Azure portal, you can select the scaling mode.



The screenshot shows the 'Create a virtual machine scale set' page in the Azure portal. The breadcrumb navigation at the top reads 'Home > Virtual machine scale sets >'. The main heading is 'Create a virtual machine scale set' followed by three dots. Below this, the 'Scaling' section is visible. It includes a 'Scaling mode' label with an information icon. There are three radio button options: 'Manually update the capacity: Maintain a fixed amount of instances.' (which is selected), 'Autoscaling: Scaling based on a CPU metric, on any schedule.', and 'No scaling profile: manual attach virtual machines after deployment'. Below these is a purple icon with a plus sign and the text 'Improve your availability by selecting multiple zones'. At the bottom, there is an 'Instance count' field with a red asterisk and an information icon, containing the value '2' and a green checkmark. Below the field is a link that says 'Configure scaling options'.

Scaling mode

- Manually update the capacity: You can manually update the capacity and maintain a fixed instances count. Set the Instance count to the number of virtual machines in the scale set (0 - 1000). Configure the [Scale-in policy](#) which is the order virtual machines are selected for deletion. For example, you could balance across zones and then delete the virtual machine with the highest instance ID.

- **Autoscaling:** You can autoscale the capacity on any schedule, based on any metrics. Specify the maximum number of virtual machines that can be available when autoscaling is applied on your implementation.

Configure autoscaling

Autoscaling is based on a scaling condition.

Add a scaling condition

×

Scale mode

☐ **Manually update the capacity:** Scaling based on a CPU metric, on any schedule

☒ **Autoscaling:** Scaling based on a CPU metric, on any schedule

Default instance count * ⓘ

Instance limit

Minimum * ⓘ

Maximum * ⓘ

Scale out

CPU threshold greater than * ⓘ

Increase instance count by * ⓘ

Scale in

CPU threshold less than * ⓘ

Decrease instance count by * ⓘ

Query duration

Minutes * ⓘ

The engine will query CPU usage for the past 60 minutes before executing the scaling to avoid reacting to transient spikes.

Schedule

Schedule type *

☒ Specify start/end dates

☐ Repeat specific days

- **Default instance count.** The initial number of virtual machines deployed in this scale set (0-1000).

- Instance limit. The minimum instance count you want this condition to scale down to. The maximum instance count you want this condition to scale up to.
- Scale out. The CPU usage percentage threshold for triggering the scale-out autoscale rule. The number of instances to scale out by.
- Scale in. The CPU usage percentage threshold for triggering the scale-in autoscale rule. The number of instances to scale in by.
- Query duration: This duration is the time the Autoscale engine looks back for the metric usage average. This look back is to allow your metric to stabilize.
- Schedule: Specify the start and end dates. You can also repeat the schedule on specific days.